Confluence Offline Workshop

FEBRUARY 12TH, 2026

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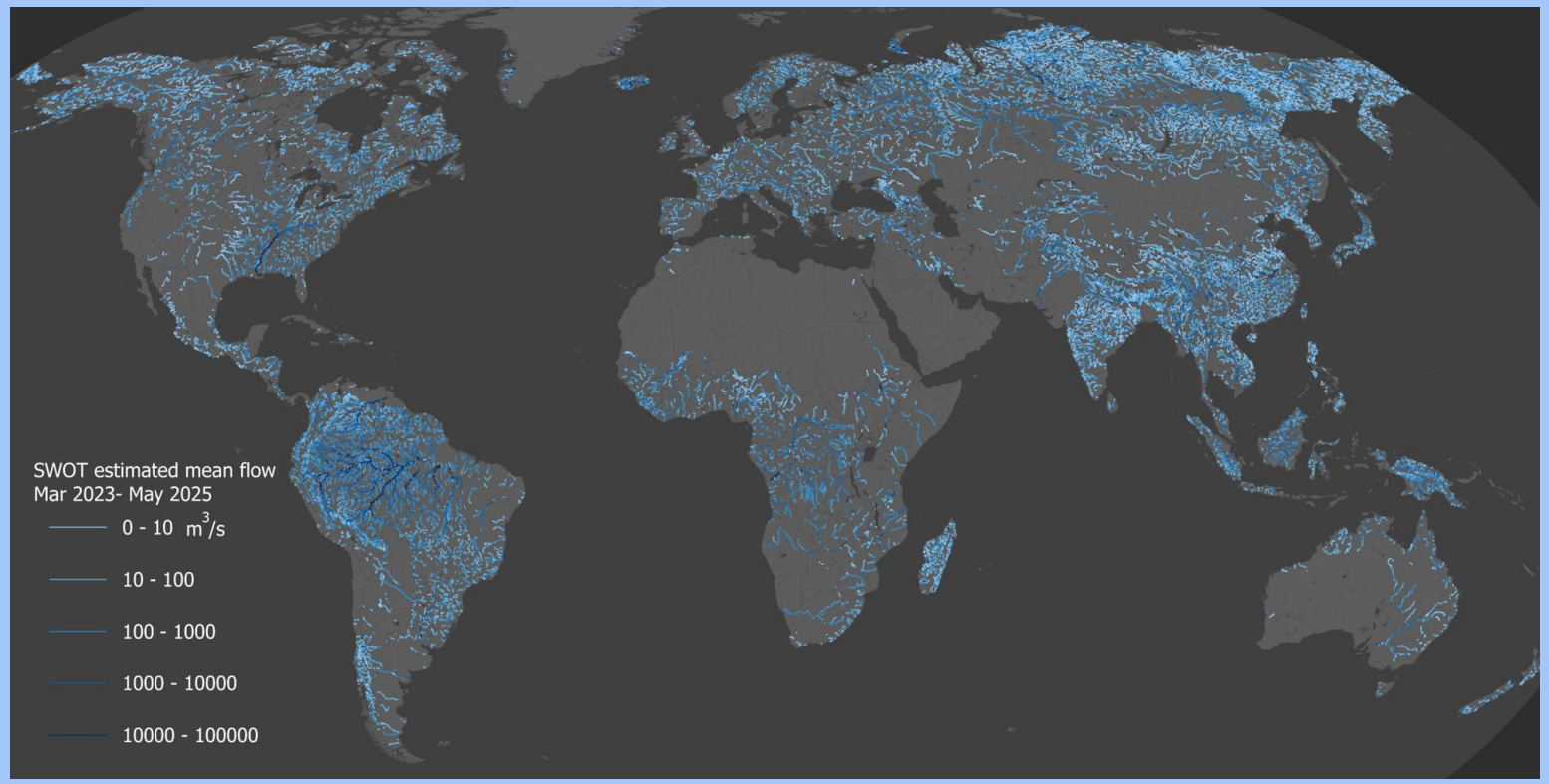
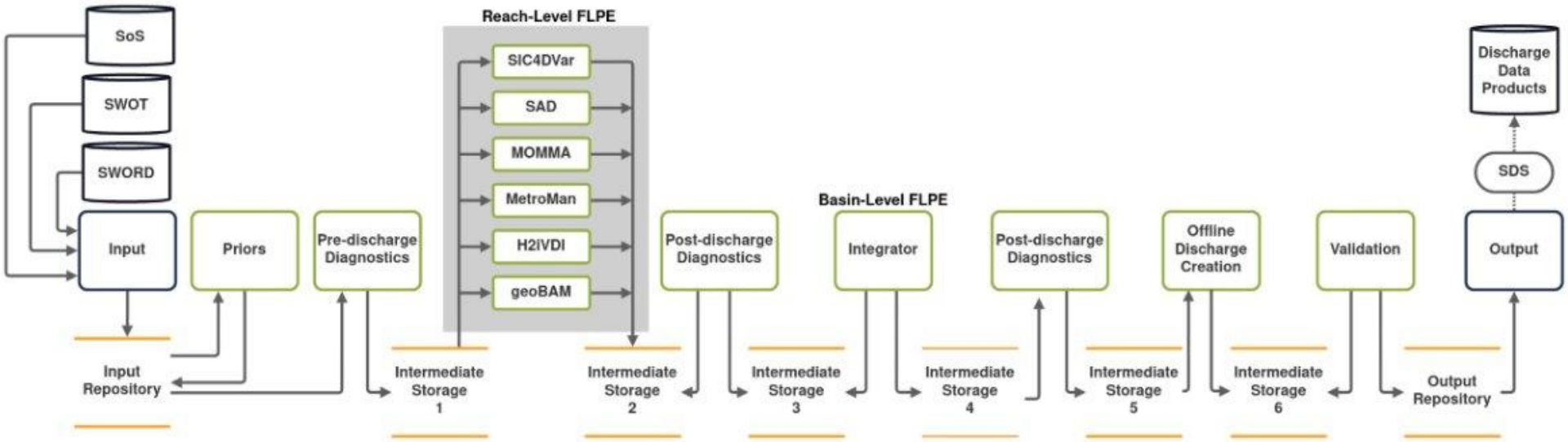


Overview

- Introduction
- Confluence Offline Structure
- Confluence Offline QuickStart
- Confluence Offline Example

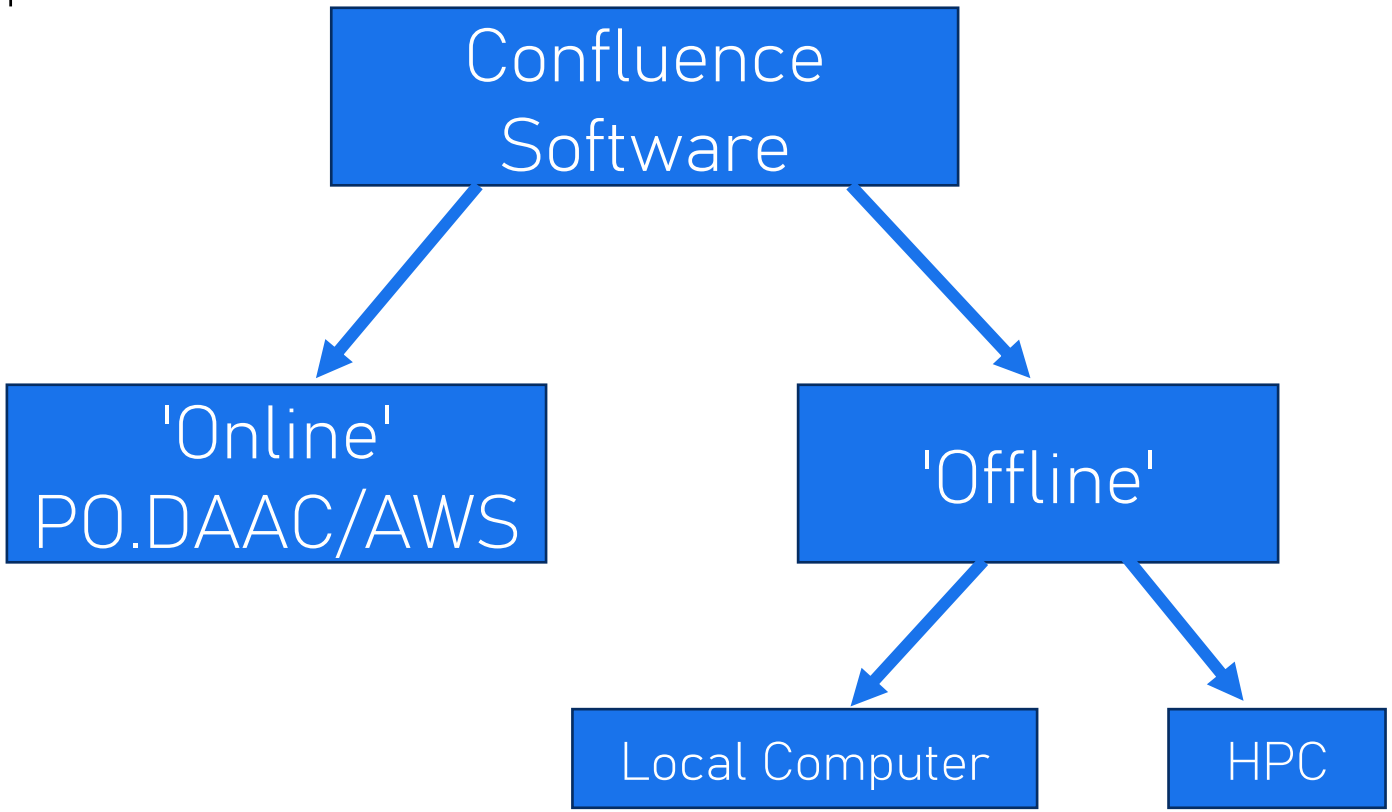


Introduction: The Confluence Framework

- The software behind the SWOT discharge product

Two approaches:

1. 'Online' – hosted by PO.DAAC
   - run on AWS
2. 'Offline' – hosted by individual
   - run on local computer or HPC



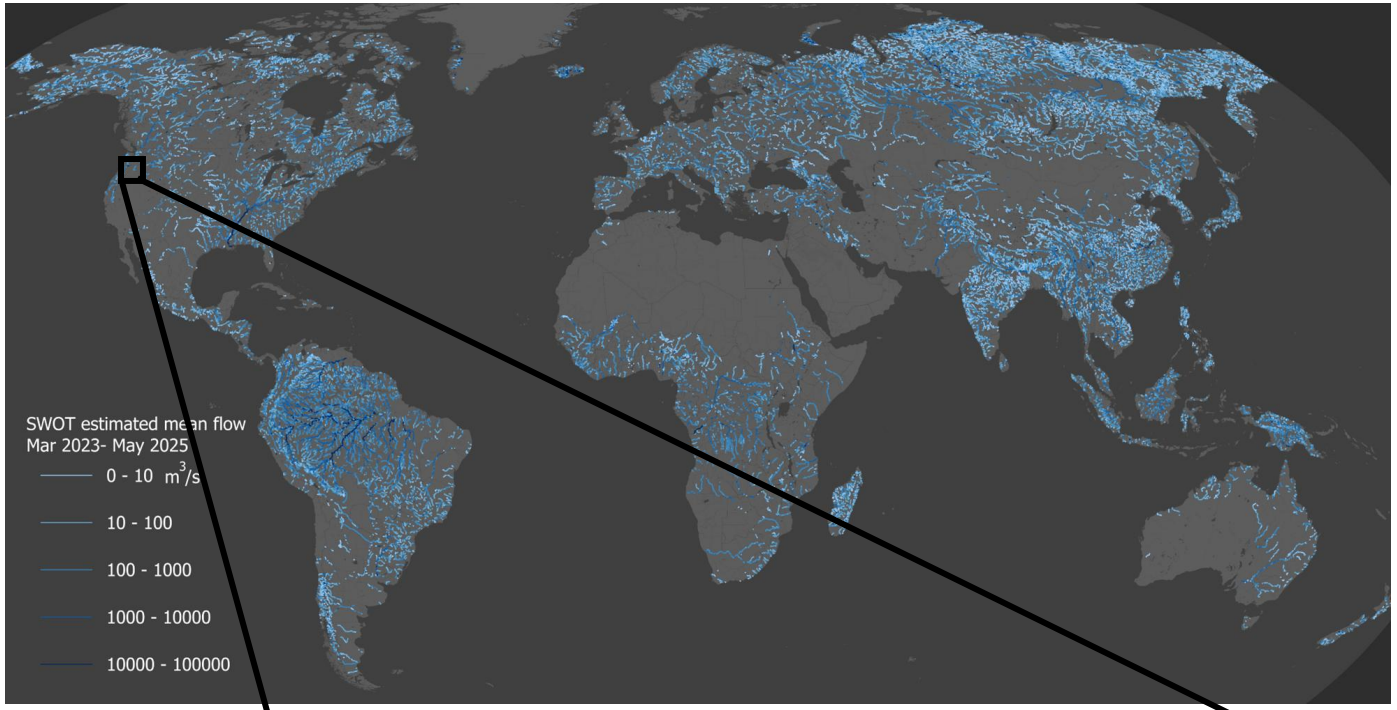
- Modular and Containerized

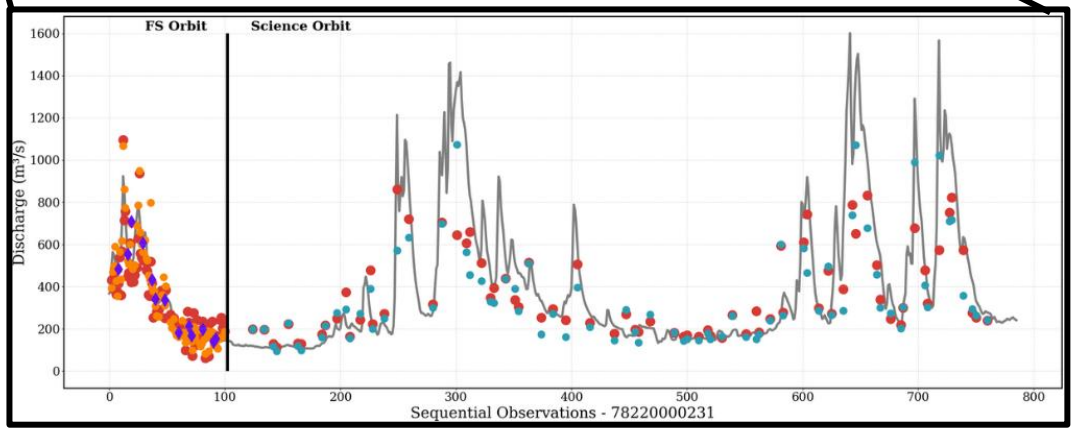
GitHub docker

- A 'living' framework – always improving thanks to you!

How Can I Use Confluence Offline?

- Individual module testing and improvement
- Access supporting data
- Customize inputs/outputs
- Incorporate latest improvements or modules
- Compute the latest results at convenience
- Ask scientific questions across scales





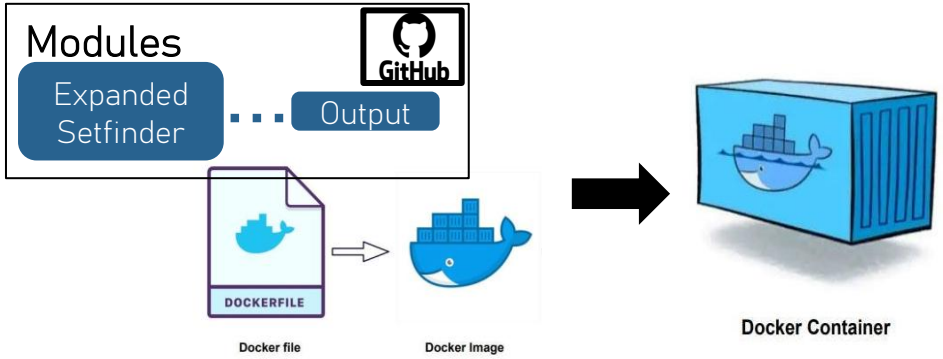
Confluence Offline Structure

At its most simple, the components of Confluence offline are:

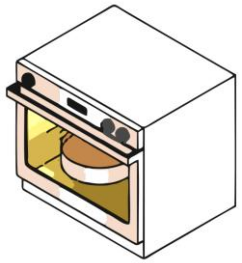
1. The pan
   - Pre-loaded prior information and directory structure

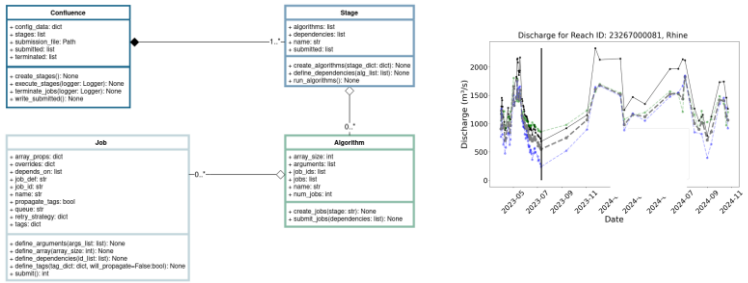
```
confluence_empty/
├── empty_mnt
│   ├── diagnostics
│   ├── flpe
│   ├── input
│   ├── logs
│   ├── mnt_dirs.sh
│   ├── moi
│   ├── offline
│   ├── output
│   └── validation
├── report
└── sif
```

2. Ingredients
   - Modules (pre-prepared using Docker)

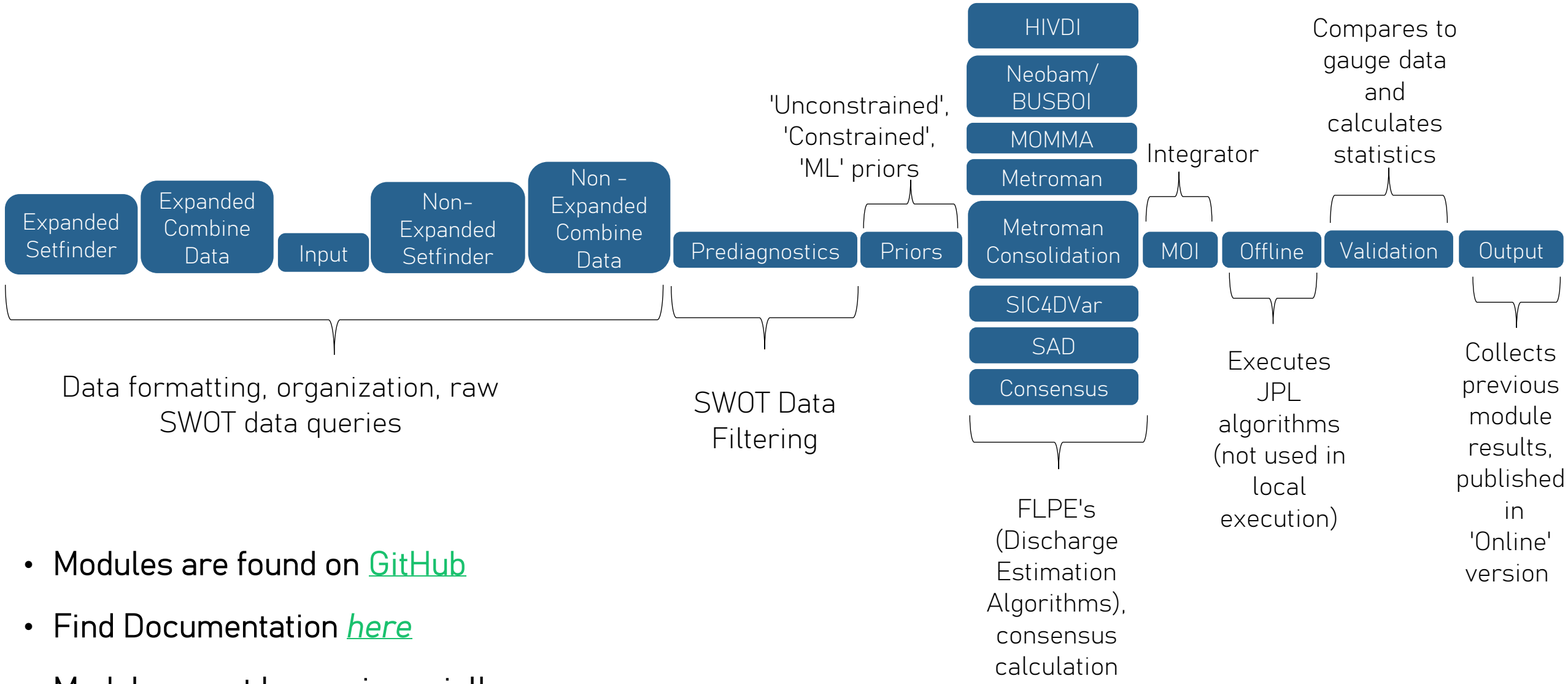


3. Assembly and Baking
   - Helper scripts (run by the user)





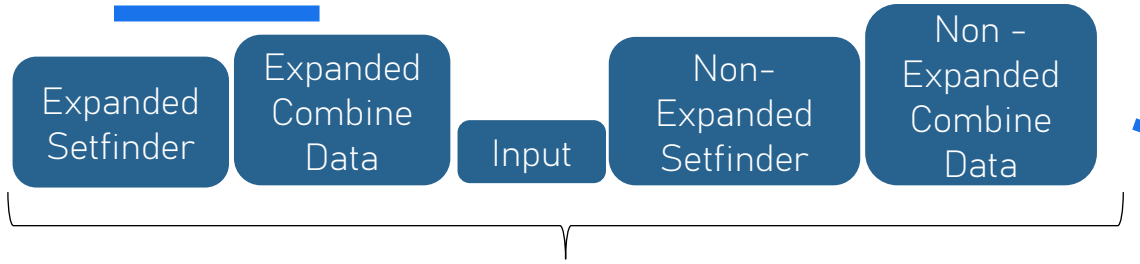
Discharge Confluence Modules* (as of Feb 2026)



- Modules are found on GitHub
- Find Documentation *here*
- Modules must be run in serial!

*Other modules such as LakeFlow or SSC are not covered in this workshop

Modules - Inputs



Data formatting, organization, raw SWOT data queries

Module	Git Branch	Number of Jobs / Reaches	Description
Expanded Setfinder (setfinder)	main	6	Creates sets (groups of connected reaches) starting with your reaches of interest and looking up and down the river
Expanded Combine Data (combine_data)	main	1	Combines the files generated in the setfinder into continent level data
Input	input_D_products	Number of Reaches	Pulls reach data from hydrocron and stores them in netcdfs, outputs to /mnt/input/swot
Non-Expanded Setfinder (setfinder)	main	6	Creates sets (groups of connected reaches) only using the reaches that were pulled successfully using Input
Non-Expanded Combine Data (combine_data)	main	1	Combines files generated in the setfinder into continent level data; OVERWRITES continent.json

```
confluence_empty/
├── empty_mnt
│   ├── diagnostics
│   │   ├── postdiagnostics
│   │   └── prediagnostics
│   ├── flpe
│   │   ├── geobam
│   │   ├── hivdi
│   │   ├── metroman
│   │   ├── momma
│   │   ├── sad
│   │   └── sic4dvar
│   ├── input
│   │   ├── 38feeef698ca484b907b7b3eb84ad05b
│   │   ├── array_check.py
│   │   ├── cont.org
│   │   ├── continent.json
│   │   ├── etc
│   │   ├── gage
│   │   ├── reaches_of_interest.json
│   │   ├── sos
│   │   ├── sword
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```

Modules – Prediagnostics and Priors

Prediagnostics | Priors

SWOT Data Filtering

'Unconstrained', 'Constrained', 'ML' priors

Unconstrained priors: WBM (NOT calibrated to gauges, Cohen et al. 2014)

Contrained priors: GRADES (calibrated to gauges, subset of gauges used for calibration and validation within confluence)

Module	Git Branch	Number of Jobs / Reaches	Description
Prediagnostics	main	Number of Reaches	Filters reach data netcdfs based on a series of bitwise filters and outlier detectors. OVERWRITES SWORD NETCDFS
Priors	main*	6	Pulls gauge data from external gauge agencies and builds the prior database (Priors SoS) – constrained and unconstrained

*ML priors are currently generated offline and already included in empty_mnt; updates to prior module in development

```
confluence_empty/
├── empty_mnt
│   ├── diagnostics
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│   │   └── prediagnostics
│   ├── flpe
│   │   ├── geobam
│   │   ├── hivdi
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```

Modules cont.

HIVDI

Neobam/ BUSBOI

MOMMA

Metroman

Metroman Consolidation

SIC4DVar

SAD

Consensus

FLPE's (Discharge Estimation Algorithms), consensus calculation

Module	Git Branch	Number of Jobs / Reaches	Description
Metroman	main	Number of Sets in `metrosets.json`	Runs the metroman FLPE algorithm, outputs to `/mnt/flpe/metroman/sets`
Metroman Consolidation	main	Number of Reaches	Takes the set level results of metroman and turns them into individual files, outputs to `/mnt/flpe/metroman`
Momma, BUSBOI, SAD, Sic4dvar, H2ivdi, consensus	main	Number of Reaches	Runs the corresponding FLPE algorithm

```
confluence_empty/
├── empty_mnt
│   ├── diagnostics
│   │   ├── postdiagnostics
│   │   └── prediagnostics
│   ├── flpe
│   │   ├── geobam
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│   ├── input
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Modules cont.

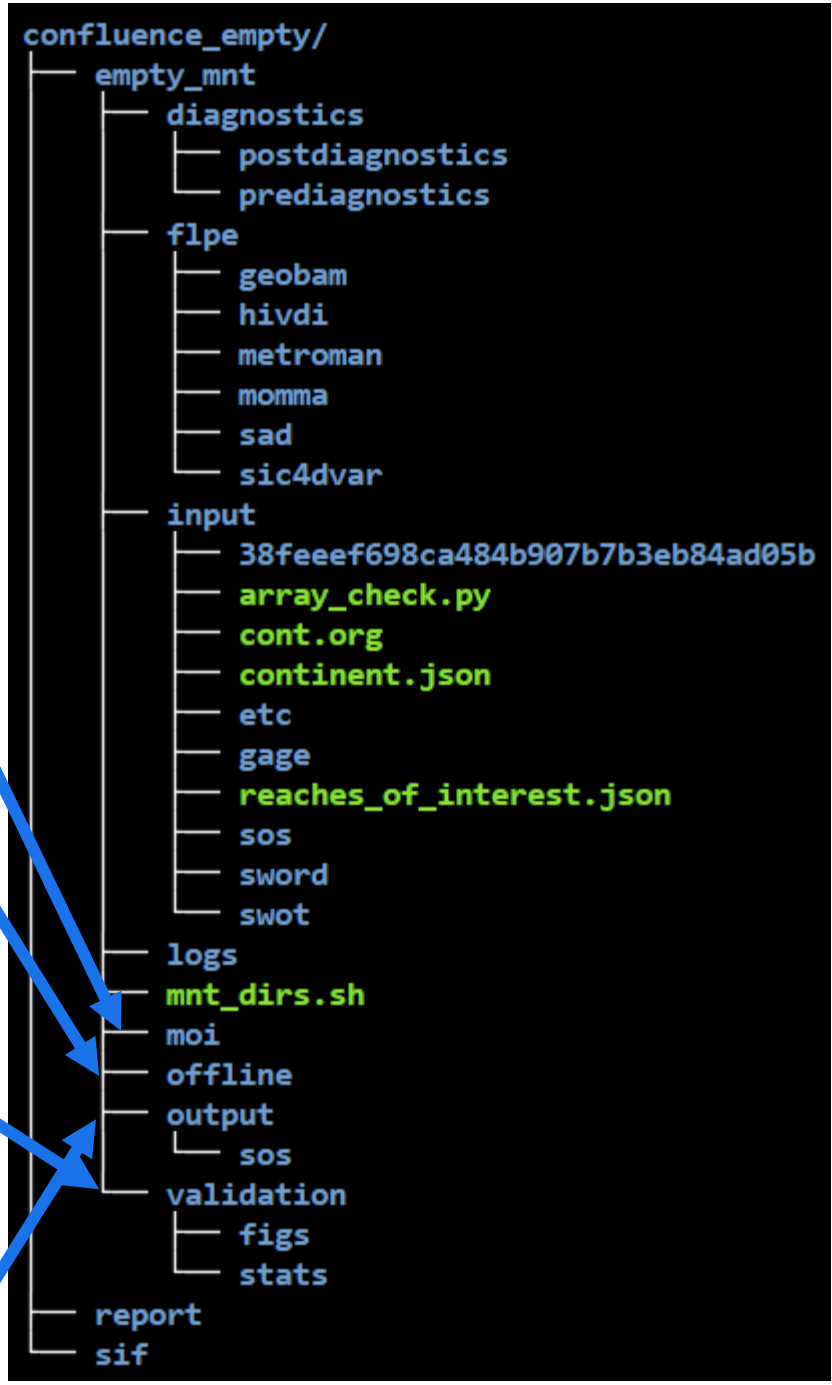
MOI | Offline | Validation | Output

MOI*

- Ensures flow conservation at river confluences
- Mass balance between rivers and lakes

*Undergoing v17 development

Module	Git Branch	Number of Jobs / Reaches	Description
MOI*	main	Number of basins in `basins.json`	Combines FLPE algorithm results
Offline (offline-discharge-data-product-creation)	main	Number of Reaches	Runs NASA SDS's discharge algorithm
Validation	main	Number of Reaches	If there is a validation gauge on the reach then summary stats are produced. (All gauges are validation in unconstrained runs)
Output	add-sword-version	6	Outputs results netcdf files that store all previous results data, outputs to `/mnt/output/sos`



Error Reporting and misc.

- Postdiagonstics – reach and discharge quality checks (under development)
- Logs - Output, warnings, errors for each reach/job (LOCAL)
- Report and Log – Output, warnings, errors for each reach/job and SLURM submission (HPC)
- sh_scripts – Contain python and bash scripts to run modules (LOCAL, HPC)
- Sif – Container image storage directory (.sif/.simg; HPC)

```
./confluence_empty
├── empty_mnt
│   ├── diagnostics
│   │   ├── postdiagnostics
│   │   └── prediagnostics
│   ├── flpe
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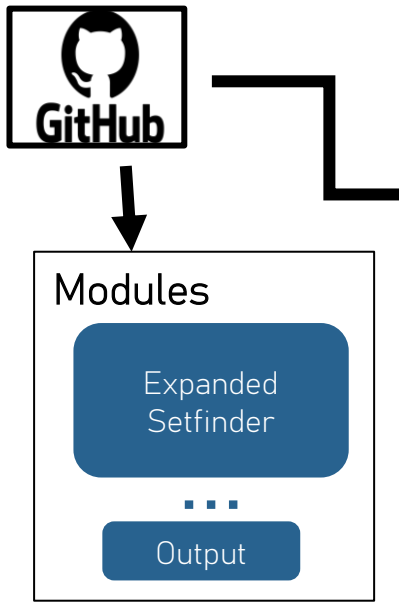
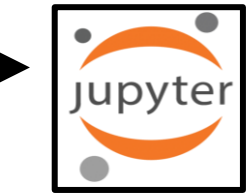
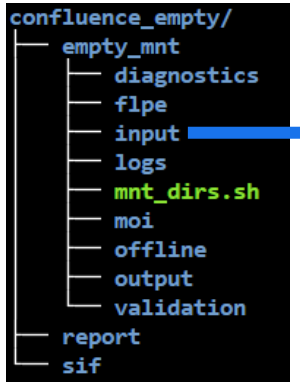
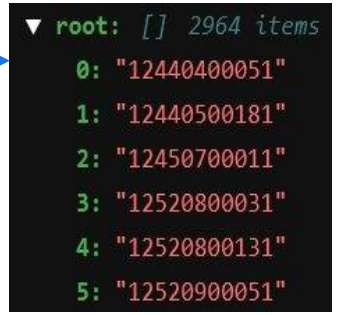
Confluence Offline Quick Start









5. Run modules (shell script commands)

1. **Fork** GitHub repositories (modules) and starter notebooks from SWOT-Confluence

2. **Download** and use Confluence local or HPC helper scripts to...

3. ...**Build and Push** modules to Docker and...

3. ...**Download** pre-populated directory (/mnt) with discharge estimation inputs and empty results folder

4. **Edit** reaches_of_interest.js on file

- Guidance, Quick Start notebooks, empty /mnt are found at [run-confluence-locally]
- Reaches file found in pre-loaded /input/

Quick Start: Summary of Inputs and Outputs

Inputs (*source*)

- Docker
- GitHub account
- Terminal and/or IDE (*i.e. Linux/VSCode*)
- Sufficient local/cloud memory (6 GB min.)
- Discharge estimation inputs (*downloaded)*
  - File structure (*/mnt*, *downloaded*)
  - SWORD (*input module*)
  - Priors (SoS, *input module*)
  - SWOT data (*downloaded during local run*, *queried from Hydrocron*)
- Modules and reaches of interest (*downloaded, SWORD*)

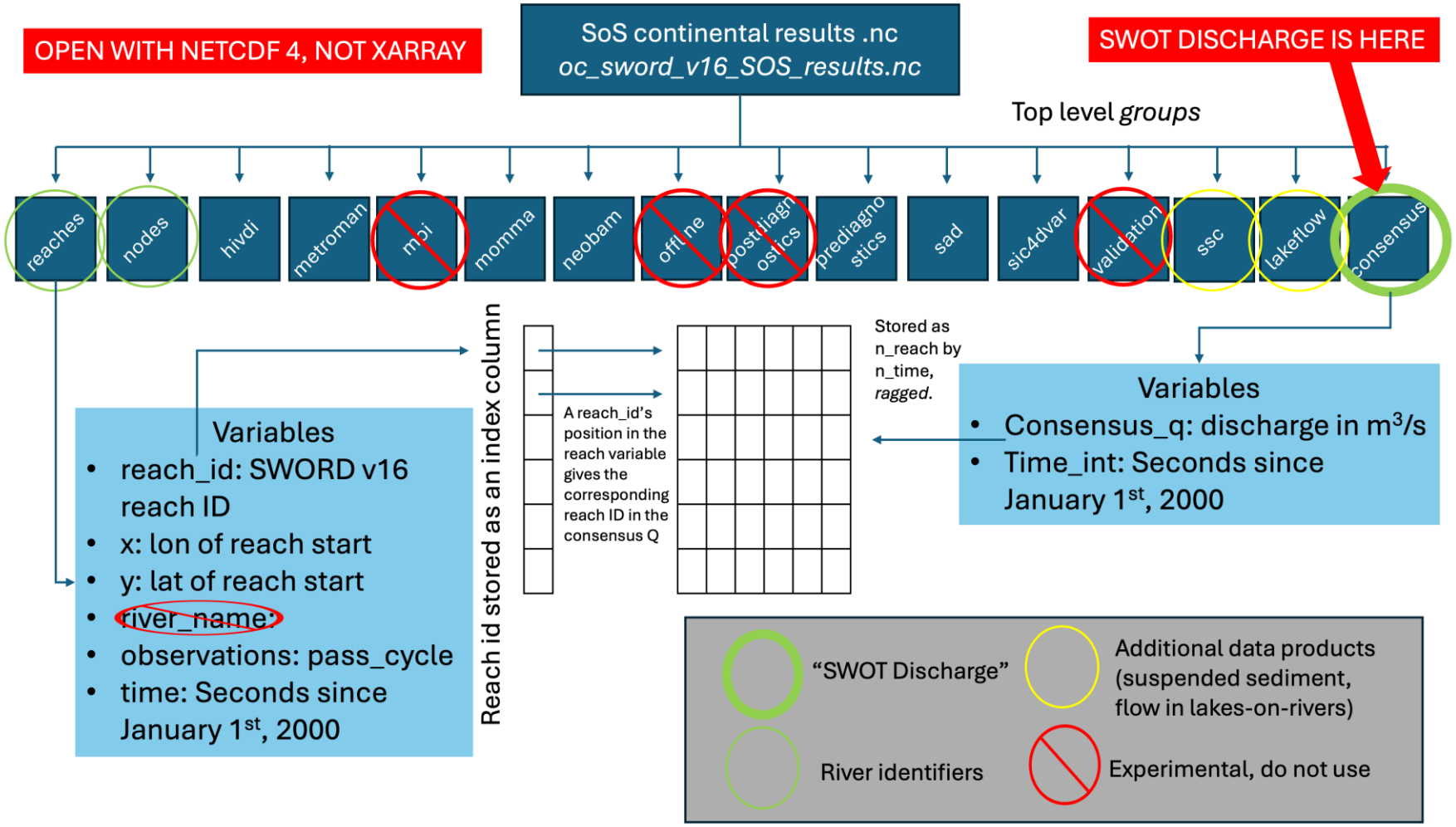
Outputs

- Discharge (m^3/s)
  - Organized as reach x time
  - File format: *.nc extension
  - Organized in 'output' module
  - Accompanying prior/algorithm data
- All accompanying Confluence pipeline module outputs
  - Usually <reach>_<module>.nc OR <continent>_<results/priors>.nc
- Error logs

Confluence Offline Example

Confluence Offline is already ahead of v1.0

1. Updated to Version D and SWORDv17
2. Updated output module to include prediagnostics and priors if needed (very large files)
3. Capacity to run H2ivdi (*included in consensus, not yet in output*)



Confluence on an HPC

Why the HPC?

- Fast and at scale (regional-global) applications
- More memory, larger analyses

Differences from Local:

- GitHub or DockerHub for module containers
- Singularity/Apptainer to build images in HPC
- Parallelized

Modifying Confluence Offline

Create Reference Confluence run
(i.e. just through input or prediagnostics)

Module Testing and Development

Local Terminal/IDE OR Docker

- Create or clone module, fork, new branch
- Alter module or scripts
  - Alter module to point to local scripts and outputs outside mnt
- Create a .venv to run module(s) of interest
  - inputs point to base mnt
  - output to new directory (symlink (linux) or alter docker command)

Scaling and Confluence Incorporation

Docker

- Create or clone module, fork, new branch
- Alter module or scripts
- Create dockerfiles and docker command
- Build and test containerized
- Run confluence and talk with the DAWG

Conclusions and Best Practices

1. Confluence is a 'living' framework
   - SWOT data, FLPE modules, SWORD, priors all change frequently
2. GitHub/Docker version control is essential
   - Fork, clone, branch, and detailed commits (Docker tags can help!)
   - Zip or delete unused Confluence directories/subdirectories to save space
3. Running through prediagnostics overwrites input and continents.json: (must re-run all modules if changing pre-algorithm or reaches of interest)
4. Confluence uses more reaches than specified in reaches_of_interest.json
   - For some algorithms, discharge is calculated on sets of reaches surrounding each reach of interest which usually increases the total reaches calculated
5. Need the latest updates on SWOT Discharge? Do you want to include your changes into Confluence?
   - Attend the bi-weekly DAWG meetings

Collected Resources

- GitHub SWOT-Confluence
- DISCHARGE :
  - Product Description Document (PDD)
  - SWOT DATASETS
  - HANDBOOK (General)
  - L4 Product 1.0, Quick User Guide
  - Video tutorials to work with the data!
- GET SWOT DATA: PO.DAAC, Hydrocron (check version using this tool too!)
- VISUALIZE DISCHARGE: PO.DAAC Cookbook
- DISCHARGE DASHBOARDS: WISP, CUAHSI Hydroshare
- SWORD: SWORD Explorer, Zenodo
- EVENTS: SWOT Significant Events

References

- Confluence Software: Tebaldi et al. Confluence Software (in review), *Environmental Modelling and Software*
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- GRADES: Lin, P. R., Pan, M., Beck, H. E., Yang, Y., Yamazaki, D., Frasson, R., et al. (2019). Global reconstruction of naturalized river flows at 2.94 million reaches. *Water Resources Research*, 55(8), 6499–6516. https://doi.org/10.1029/2019wr025287
- SWOT Q PAPER: Andreadis, K. M., Coss, S. P., Durand, M., Gleason, C. J., Simmons, T. T., Tebaldi, N., … & Yadav, B. (2025). A first look at river discharge estimation from SWOT satellite observations. *Geophysical Research Letters*, *52*(9), e2024GL114185.

Questions?

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Virtual Environment Quick Start

Example: running Sic4dvar

Fork and Clone Sic4dvar from SWOT-Confluence

- You can create a virtual env and install the mandatory libs:
  - python3 -m venv your_env_name
  - source your_env_name/bin/activate
  - pip install --upgrade pip
  - pip install -r requirements.txt (the requirements.txt should be provided in the repo)
  - To load the env as a kernel: python -m ipykernel install --user --name=your_env_name
- Run sic4dvar locally, without building the docker
- python3 sic4dvar.py --json_path input/reaches.json --output_dir sic4dvar --log_dir sic4dvar_log --swot_dir input/swot/ --sos_dir input/sos/ --sword_dir input/sword/ -l INFO -p

Docker Command Quick Start

'expanded_setfinder': f'docker run -v {mnt_dir}/input:/data {docker_username}/setfinder:{custom_tag_name} -r reaches_of_interest.json -c continent.json -e -s 17 -o /data -n /data -a MetroMan HiVDI SIC -i {{index}}',

Base docker command to start container

Map a directory from host machine to container

Source directory on host, /data is targe directory in container

Container image name

Arguments (based on module scripts)

Index of reach/continent for a loop/array job